3.2 Amino acids

There are more than 300 amino acids found in nature. However, only 20 of them are present in proteins. Each amino acid has three distinct groups

a. A primary amino group (Except for the amino acid proline, which has a secondary amino or imino group)

b. A carboxylic acid group and

c. A distinct side chain or R group.

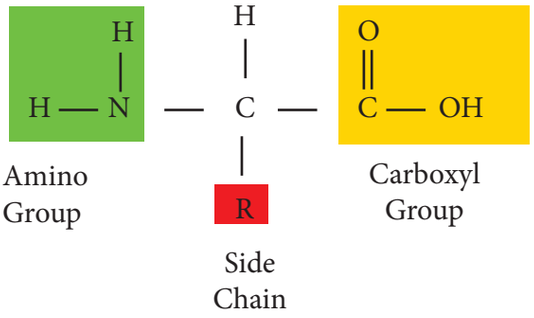

Figure 3.2 Chemical structure of an amino acid

The chemical nature of the side chain in each amino acid determines the property of that amino acid and the role it plays in a protein structure. Considering this important role of the side chain, amino acids are classified into different groups based on their side chain.

3.2.1 Amino acids with non polar side chains

These amino acids have non polar groups as their side chains and are shown in Table 3.1. These non polar side chains are similar to lipids and tend to clump together in an aqueous environment by means of a special type of interaction called hydrophobic interaction. Hence, these amino acids are also called hydrophobic amino acids. These side chain groups (are less likely to lose or gain protons) form hydrogen bond or ionic bonds.

3.2.2 Amino acids with uncharged polar side chains

The amino acids serine, threonine, tyrosine, cysteine, asparagine and glutamine have a net zero charge at neutral pH. However at alkaline pH the side chain of cysteine and tyrosine can lose a proton. The polar hydroxyl group (-OH) present in the side chains of serine, threonine and tyrosine can form a hydrogen bond. Similarly, the cysteine side chain can form disulphide bonds. The details of hydrogen bond and disulphide bonds along with their role in protein structure and stability will be explained in a later section of this unit.

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